# Audio Engineering & Software Portfolio

Audio Software, DSP & Interactive Systems — C++/JUCE, MATLAB & Python

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## 1 Professional Profile

Software developer focused on **Digital Signal Processing (DSP)** and professional audio tooling, with an ecosystem of VST3 plugins published under the **ESP** brand and distributed through a self-built web store. Experienced in modern **C++** with the **JUCE** framework, **MATLAB** and **Python** — from wavetable synthesis and real-time dynamics processing to gesture-controlled instruments with computer vision, embedded systems (ESP32, Raspberry Pi/ROS2) and applied machine learning. Prioritises *lock-free* architectures, algorithmic efficiency and clear real-time data visualisation.

## 2 Audio Plugins — C++ / JUCE

VST3 / Standalone plugins developed in C++ with JUCE 8, published under the **ESP** brand and distributed at ESP Plugin Store.

### 2.1 SYNTH/1 — Polyphonic Wavetable Synthesizer

16-voice polyphonic synthesizer with a Serum-inspired architecture: wavetable morphing, a unison engine, a step sequencer and a full effects rack.

- **Oscillator:** 4 band-limited wavetables (64 harmonics) with continuous morphing and 4 phase-warp modes (Sync, Bend, PWM); real-time waveform visualiser.
- **Voice architecture:** up to 8-oscillator unison with constant-power detune and stereo spread; Poly/Mono/Legato voice modes with per-voice multiplicative glide; chain: oscillator → pan → tanh drive → SVT filter → ADSR.
- **Modulation & sequencer:** LFO routable to cutoff and pitch with a mod-depth ring drawn around the knob; step sequencer of up to 16 steps with internal clock or DAW triggering, fully serialised in the plugin state.
- **FX & EQ:** chorus, phaser, tempo-synced delay, tanh saturation with 2× oversampling and reverb; **interactive** 3-band EQ (drag the bands directly on the frequency response) with a live FFT spectrum overlay.
- **Engineering:** lock-free audio↔UI communication (atomics and ring buffers), XML presets via APVTS, 7 colour themes with a custom *LookAndFeel*.
- **Repository:** GitHub – Synth 1.0


Figure 1: SYNTH/1: main tab with wavetable visualiser, filter and ADSR.

## 2.2 SPECTRUM — Real-Time Spectrum Analyzer

High-precision spectral analysis tool operating in real time with optimised latency.

- **DSP engine (FFT):** 2048-point FFT with Hann windowing; audio is pulled through *thread-safe* FIFO structures, avoiding any blocking of the high-priority audio thread.
- **Rendering:** spectral curve with Bézier interpolation and moving-average smoothing, logarithmic frequency mapping (20 Hz–20 kHz), 60 FPS refresh with a minimal CPU footprint.
- **Metering:** analog-style VU meters in dBFS with smooth falloff ballistics, stereo panning indicator and FREEZE function; multiple views (spectrogram, 3D waterfall, waveform) and colour themes.
- **Repository:** GitHub – Spectrum Analyzer

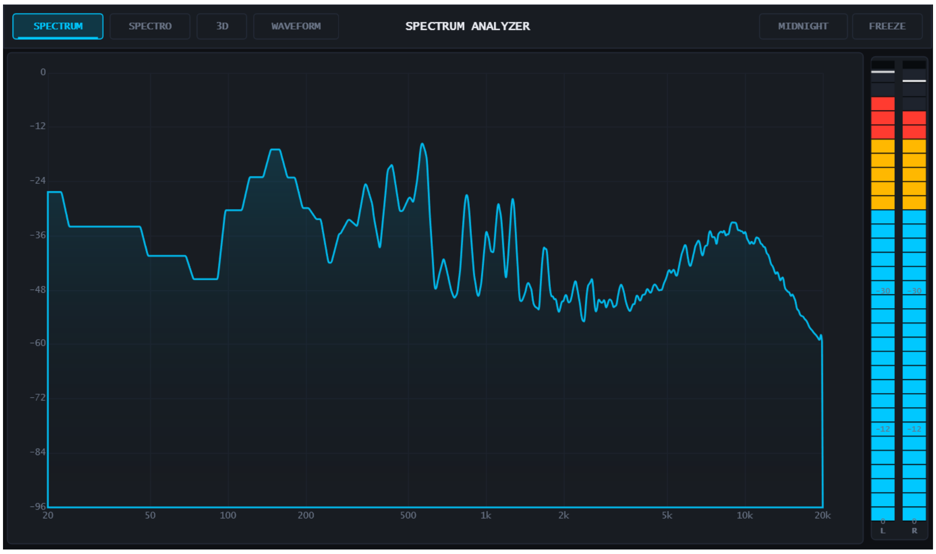


Figure 2: SPECTRUM showing the smoothed frequency response and VU meters.

## 2.3 VERTEX — Dynamic Range Compressor

Professional-grade dynamics processor with exhaustive control over the signal envelope and instant visual feedback of the gain reduction.

- **Architecture & APVTS:** full *AudioProcessorValueTreeState* integration for safe communication between the processing thread and the GUI, with persistent preset storage.
- **Dynamics processing:** gain computation with attack and release envelopes parameterisable in real time; Threshold, Ratio, Attack, Release and Makeup Gain controls.
- **Visual feedback:** dynamic transfer-curve (*knee*) display that updates with the user's parameters, complemented by asynchronous peak meters.
- **Repository:** GitHub – VERTEX

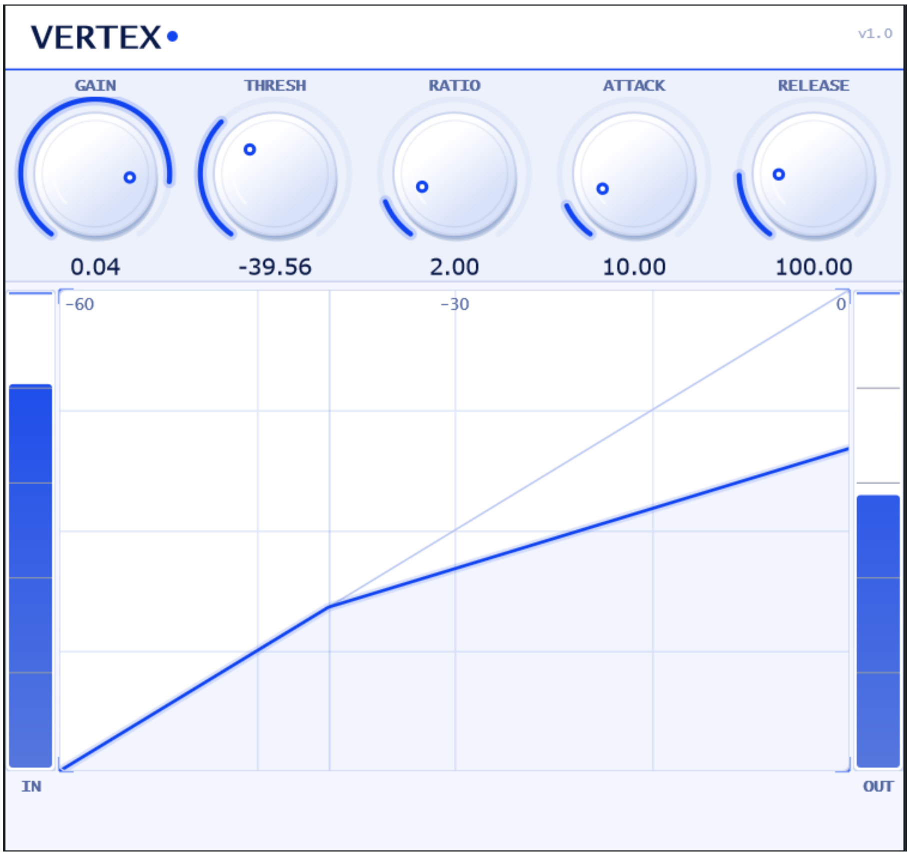


Figure 3: VERTEX: transfer curve and gain-reduction metering.

### 2.4 ESP-L1 — Brick-Wall Limiter

Instant-attack limiter with built-in spectral analysis to monitor the effect of the processing on the frequency content.

- **Limiting:** per-sample envelope detector with exponential decay (10–1000 ms release with perceptual skew), −60 to 0 dB threshold and post-limiting output gain.
- **Pre/post analysis:** two independent FFT pipelines (2048-point, Hann) draw the spectrum before and after limiting, overlaid, alongside VU and gain-reduction meters.
- **Threading:** audio/UI state shared through atomics; the UI computes FFTs and ballistics at 30 Hz without touching the audio thread.
- **Repository:** GitHub – ESP-L1


Figure 4: ESP-L1: pre/post limiting spectrum and gain-reduction meter.

### 2.5 MEGACRUSHER — Distortion, Saturation & Bit-Crusher

Distortion plugin with three saturation algorithms, bit-depth reduction and an animated interface that reacts to the signal.

- **Saturation:** SOFT (tanh), HARD (hard-clip) and FOLD (wave-folder that mirrors the signal back into range, generating complex harmonics) modes; $1\times$ to $40\times$ drive stage.
- **Timbre:** 16-bit down to 2-bit bit-crusher, one-pole tilt filter for tone control and dry/wet mix for parallel distortion.
- **Visualisation:** live transfer-curve display, stereo peak-hold VU with colour-coded segments and an animated UI (ember particles and flames reacting to the Drive level).
- **Repository:** GitHub – MegaCrusher


Figure 5: MEGACRUSHER: live saturation curve and stereo VU.

### 2.6 Basic Oscilator — Oscillator Generator

Three independent oscillators with waveform selection (sine, saw, square) built on `juce::dsp`; first project of the JUCE ecosystem and the experimental foundation for SYNTH/1.

### 2.7 ESP Plugin Store — Web Distribution

Self-built web store distributing the ESP-brand plugins, deployed at esp-plugin-store.vercel.app.

- **Frontend:** self-contained React 18 SPA (no build step); seeded generative waveform art per plugin, screenshot carousel and a download modal with OS and format selection.
- **Distribution:** VST3 binaries packaged under the GPL v3 licence, SEO sitemap and continuous deployment on Vercel.

## 3 Signal Processing in MATLAB

### 3.1 5-Band Parametric Equalizer

5-band parametric equalizer programmed entirely in MATLAB using object-oriented programming, focused on precise audio-signal manipulation and interactive visualisation of the frequency response.

- **Filters (Biquad):** second-order IIR filters (Direct Form II) for Peaking, High-Pass and Low-Pass stages; real-time computation of the complex transfer function $H(z)$ to derive the magnitude response.
- **Audio engine:** optimised processing loop guaranteeing smooth coefficient updates, preventing digital artefacts during dynamic manipulation of Gain, Freq and Q.

- **Analysis & UI:** interface (*uifigure*) built 100 % in code with logarithmic 20 Hz–20 kHz mapping on a decibel scale, plus preset system (Flat/Rock/Jazz/Vocal/Bass/Classical) and spectrogram and pole–zero analysis views.
- **Repository:** GitHub – Parametric Equalizer


Figure 6: Parametric EQ during playback: draggable band handles over the live input spectrum.

# 4 Gestural Instruments — Python & Computer Vision

## 4.1 Aetheric Geometry — Real-Time Gesture Instrument

Musical instrument played with bare hands: a webcam feeds MediaPipe hand landmark detection and the geometry of the hands is translated into additive synthesis with just intonation.

- **Gesture pipeline:** gesture detectors (pinch, kiss, pyramid, cross, prayer, open palm) on top of a state machine (IDLE / THREAD / POLYGON) with debouncing and robust hand assignment; waveform selection by counting extended fingers.
- **Synthesis engine:** additive oscillators with just-intonation ratios, asymmetric envelope, IIR low-pass filter mapped to hand height, Schroeder reverb (4 comb filters) and tremolo driven by the polygon's aspect ratio.
- **Product:** built-in WAV recording, optional MIDI output, packaged as an executable (PyInstaller) and a *pytest* suite that requires no camera or audio hardware.
- **Repository:** GitHub – Aetheric Geometry

## 4.2 AirStems — Gestural Stem Remixing (Musicathon by Musixmatch Pro, 2026 — LALAL.AI Special Prize)

"Conduct and remix a song's stems with your bare hands": source separation + synced lyrics + gestural control over a live DSP engine.

**Won the LALAL.AI Special Prize** at this global online hackathon; since then **featured in a published LALAL.AI interview** and invited to write a **technical guest article** for the LALAL.AI blog and Dev.to.

- **Stems & lyrics:** vocals/drums/bass/other separation via the **LALAL.AI** API (with a local Demucs fallback) and **Musixmatch** time-synced lyrics scrolling karaoke-style.

- **Gestural control:** two-hand MediaPipe tracking — right-hand fingers toggle each stem (with hysteresis), left-hand height drives a low-pass filter and finger spread the reverb; a fist performs a drop.
- **Real-time engine:** stereo multi-stem mixer on a `sounddevice` callback with gain changes quantised to the beat grid (librosa on the drum stem) and click-free per-block ramps.

### 4.3 AirPiano — Webcam Air Piano

8-note piano (plus a drum-kit variant) played in the air: frame-difference motion detection with OpenCV over on-screen key zones and sample triggering with pygame; motion-threshold calibration for the room's lighting.

## 5 Hardware & Embedded Systems

### 5.1 EDISA Inventory Drone — Autonomous Warehouse UAV (UPC PAE × EDISA, 2025–2026)

Autonomous drone that flies through a warehouse, builds a real-time 3D map, detects barcodes and packages, and generates a geolocated inventory — no manual operation. Team project of 14 students for the industry client EDISA; personal focus on WP2: the Ground Control Station and drone-control stack.

- **Onboard system:** Pixhawk/ArduCopter flight controller plus Raspberry Pi 5 running ROS2 Jazzy; Unitree 4D LiDAR feeding *Point-LIO* SLAM (∼800,000 points per scan), Pi Camera, barcode detection (OpenCV, pyzbar, YOLOv8) with each scan tagged with its SLAM x/y/z position.
- **Ground Control Station:** cross-platform Electron app — telemetry/arming dashboard, artificial-horizon navigation, 3D point-cloud SLAM viewer and an inventory tab backed by SQLite with CSV/Excel export; talks to the drone over rosbridge WebSocket and MJPEG streaming, with no ROS install required on the laptop.
- **Autonomy:** mission-coordinator state machine (brain node) driving waypoint navigation over MAVLink v2; physics-based flight simulator and Monte Carlo mission simulations for validation.

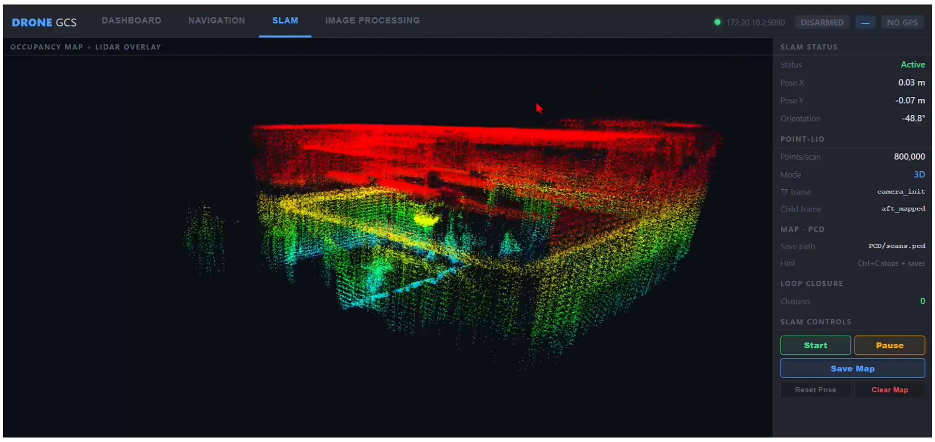


Figure 7: EDISA drone GCS: live 3D LiDAR point cloud in the SLAM viewer.

### 5.2 LaserHarp — Laser Harp

Musical instrument in which interrupting laser beams triggers the notes; the project covers both the electronics design and the control software.

- **Hardware:** Fritzing protoboard and PCB designs iterated v1 → v3 with ToF (*Time-of-Flight*) distance sensors and a Raspberry Pi control layer; ToF synthesizer prototype ("ToF-Synth").
- **Software:** Python/Tkinter simulator of the instrument's 16-character LCD menu (state machine for navigating and editing ADSR, 3-band EQ, effects and MIDI configuration) with Unicode-rendered value bars.

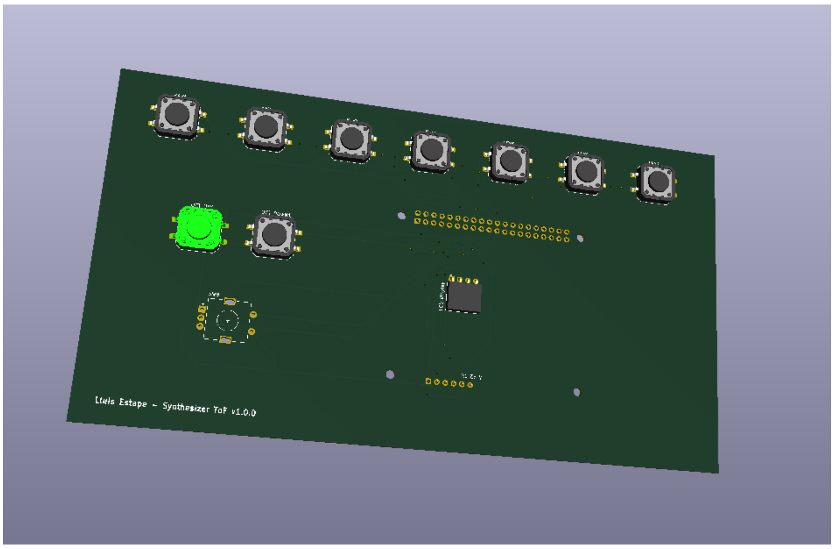


Figure 8: LaserHarp: 3D render of the ToF synthesizer PCB (v1.0.0).

### 5.3 ESP32 Cyberdeck (in progress)

Hand-built audio handheld (synth / sampler / MIDI): a sandwich of two perfboards with the controls on the walls of the central cavity.

- **3-microcontroller architecture:** ESP32-DevKitC as the brain (ILI9341 TFT display over SPI, $I^2S$ audio, $I^2C$ sensors), a second ESP32 as an $I^2C$ slave dedicated to reading 2 encoders, a slider, buttons and a D-pad, and an ESP32-CAM streaming MJPEG video over WiFi.
- **Audio chain:** $I^2S$ → PCM5102A DAC → PAM8403 amplifier → speaker; environmental sensors (AHT10, PIR) behind a TCA9548A multiplexer; 2000 mAh LiPo power with a charger/boost module.
- **Enclosure:** parametric case generator (10 bodies, display and camera windows, speaker grille, "hat" push-buttons) written in Python against the Autodesk Fusion 360 API.

## 6 Machine Learning & Other Projects

### 6.1 QTGNN v2 — Quantum Temporal Graph Neural Network

Hybrid classical–quantum model for next-day stock price prediction, documented in a full technical report.

- **Architecture:** GRU for temporal encoding of price and volume, FinBERT transformer for real-time financial news sentiment, and a *Quantum Graph Attention Network* propagating information over a stock-correlation graph through a 4-qubit variational quantum circuit.
- **Data & validation:** 10 S&P 500 stocks across 5 sectors, ~500 days of history (OHLCV via yfinance) plus macroeconomic indicators, validated with walk-forward cross-validation.

### 6.2 AetherMap — Visual Blast-Radius Review (GitLab Transcend Hackathon, June 2026)

Agent for the GitLab Duo Agent Platform that turns the *Orbit Knowledge Graph* into a blast-radius review of every merge request.

- **Analysis engine:** maps changed hunks to the definitions that contain them and walks the call graph backwards (CALLS/EXTENDS edges) to find everything transitively impacted; scores risk LOW to CRITICAL, flags the most widely-called "fragile cores" and suggests reviewers.
- **Product:** Markdown report posted as an MR comment plus an interactive force-directed impact map; packaged as an Agent Skill with an MR-triggered CI flow. 100 % standard-library Python, MIT licence, unit-test suite.

### 6.3 Portfolio Website

Personal portfolio site with component-driven development in React (JSX) and Tailwind CSS, from UX wireframes to the optimised production export. **Repository:** GitHub – Portfolio Website

## 7 Technical Skills

- **Languages & paradigms:**
  - Modern C++ (C++17/20), Python, MATLAB and JavaScript (React, Electron).
  - Object-oriented programming: modular architecture, structured class design and strict encapsulation.
- **Audio software development:**
  - **JUCE 8:** full plugin development in industry-standard formats (VST3, Standalone) and binary publishing.
  - **Real-time DSP:** IIR/FIR filters, FFT, dynamics control, wavetable/additive synthesis, oversampling and algorithmic reverberation.
  - Lock-free GUI/audio communication: atomics, FIFOs and ring buffers.
- **Computer vision & ML:**
  - OpenCV, MediaPipe (hand tracking), YOLOv8 and librosa (beat analysis).
  - Neural networks (GRU, transformers, GNN) and quantum machine learning with variational circuits.
- **Robotics, embedded & hardware:**
  - ROS2 (Jazzy), MAVLink, LiDAR SLAM (Point-LIO) and rosbridge integration.
  - ESP32 ($I^2S$, $I^2C$, SPI), circuit design in Fritzing and parametric modelling with the Autodesk Fusion 360 Python API.
- **Tools & environments:**
  - Microsoft Visual Studio 2022 (primary environment), MATLAB Editor.
  - Git and GitHub (repositories and Releases), Vercel deployment, GitLab CI, LaTeX.
  - Third-party API integration: LALAL.AI, Musixmatch, GitLab Orbit.